

ANTABENZ INJECTION

DESCRIPTION: ANTABENZ INJECTION is a colourless to almost colourless clear liquid.

COMPOSITION: Each ml contains Flumazenil 0.1mg.

PHARMACODYNAMICS: Flumazenil, an imidazobenzodiazepine, is a benzodiazepine antagonist which specifically blocks the central effects of agents acting through the benzodiazepine receptor by competitive inhibition.

Pharmacodynamics: Intravenous Flumazenil has been shown to antagonize sedation, impairment of recall, psychomotor impairment and ventilatory depression produced by benzodiazepine in healthy human volunteers. The duration and degree of reversal of benzodiazepine effects are related to the dose and plasma concentration.

PHARMACOKINETICS: Flumazenil, a weak lipophilic base, is about 50% bound to plasma proteins. Albumin accounts for two thirds of the plasma protein binding. The average elimination half-life of the drug is 53 minutes. The mean volume of distribution at steady state ($V_{ss} = 0.95 \text{ L/kg}$) is close to that of structurally related benzodiazepines and indicates tissue binding and/or partitioning of the drug.

Flumazenil is almost completely (99%) non-renal eliminated. The carboxylic acid was identified in free and conjugated form as the main metabolite in human urine. In pharmacological tests, this main metabolite was inactive as a benzodiazepine agonist or antagonist. The total plasma clearance of Flumazenil is on average 1 L/min. and can be attributed almost entirely to hepatic clearance. The low renal clearance rate suggests an effective reabsorption of the drug after glomerular filtration. When administered together with the benzodiazepines midazolam, flunitrazepam or lorazepam, the basic pharmacokinetic parameters of Flumazenil were not affected.

INDICATIONS: Flumazenil is indicated for reversal of the centrally sedative effects of benzodiazepines. It is therefore used in anesthesia and intensive care in the following indication.

In Anesthesia: Termination of general anesthesia induced and maintained with benzodiazepines in patients. Reversal of benzodiazepine sedation in short diagnostic and therapeutic procedures in both inpatients and outpatients.

In Intensive Care: Flumazenil provides a diagnostic proof of intoxication with benzodiazepines or rules out such intoxication.

As a diagnostic measure in unconsciousness of unknown origin to differentiate between involvement of benzodiazepine, other drugs or brain damage.

As specific reversal of the central effects of benzodiazepine in drug overdose (return to spontaneous respiration and consciousness in order to render intubation necessary or allow extubation).

RECOMMENDED DOSAGE: Flumazenil should be administered intravenously by an anaesthesiologist or experienced physician. The use of Flumazenil should be balanced against the risk of precipitating withdrawal symptoms. The desirability of retaining a degree of sedation in the early postoperative period should be considered.

Flumazenil should be diluted with dextrose 5% lactated Ringer's solution or Sodium Chloride 0.9% or Sodium Chloride 0.45% + dextrose 2.5% for infusion and may also be used concurrently with other resuscitative procedure. Dosage should therefore be titrated for the intended effect. Since the duration of action of some benzodiazepines may exceed that of flumazenil, repeated doses may be required if sedation recurs following awakening. In order to reduce microbial contamination hazards, infusion should be commenced as soon as practicable after preparation. Infusion should be completed within 24 hours of preparation and any residue discarded.

Reversal (anaesthesia or sedation):

The recommended initial dose is 0.2 mg administered i.v. within 15 seconds. If the desired degree of consciousness is not obtained within 60 seconds following the first i.v. administration, a second dose of 0.1 mg can be injected and this may be repeated at 60 second intervals where necessary, up to a total dose of 1 mg. The usual dose is 0.3-0.6 mg.

Individual requirement may vary considerably, depending on the dose and duration of effect of the benzodiazepine administered and patient characteristics. Rapid injection of flumazenil in patients with long-term exposure to benzodiazepines during the weeks preceding flumazenil administration may produce withdrawal symptoms and should therefore be avoided. In case of unexpected signs of withdrawal symptoms, a slow i.v. injection of 5mg diazepam or 5mg midazolam should be given.

In the Intensive Care Unit:

The recommended initial i.v. dose is 0.3 mg. If the desired degree of consciousness is not obtained within 60 seconds, Flumazenil may be injected repeatedly until the patient awakes or up to a total dose of 2 mg. If drowsiness recurs, an i.v. infusion of 0.1 - 0.4 mg/hour has been shown to be useful. The rate of the infusion should be individually adjusted up to the desired level of arousal.

If a significant improvement in consciousness and respiratory function is not obtained after repeated doses of flumazenil, a non-benzodiazepines etiology must be assumed. In the intensive care unit, in patient treated with high doses of benzodiazepines and/or for long periods of time, the individually titrated injections of flumazenil, slowly administered should not produce withdrawal symptoms. If unexpected signs of overstimulation occur, 5mg diazepam or 5mg midazolam should be given intravenously.

ROUTE OF ADMINISTRATION: For i.v. injection / i.v. infusion only.

CONTRAINDICATIONS:

Flumazenil is contra-indicated in patients with known hypersensitivity to the drug.

Flumazenil is contraindicated in patients who have been given a benzodiazepine for control of a potentially life-threatening condition (e.g. control of intracranial pressure or status epilepticus).

WARNING & PRECAUTIONS: Precautions:

Consideration should be given to the possibility of re-sedation, respiratory depression or other residual benzodiazepine effects following the use of Flumazenil. These patients should be monitored for an appropriate period based on the dose and duration of effect of the benzodiazepine employed. Although after intravenous administration of Flumazenil the patients are awake and conscious, they should be warned against engaging in hazardous occupations requiring complete mental alertness such as operating dangerous machinery or driving a motor vehicle during the first 24 hours after administration.

Flumazenil should not be used as a routine empirical means of assessing unconscious patients in settings where resuscitation equipment and expertise to deal with complications are not immediately to hand. In addition, Flumazenil may be capable of precipitating convulsions or altering cerebral blood flow in patients with head injury receiving benzodiazepines.

The use of Flumazenil is not recommended in epileptic patients who have been receiving benzodiazepine treatment for a prolonged period. Although Flumazenil exerts a slight intrinsic anticonvulsant effect, its abrupt suppression of the protective effect of a benzodiazepine agonist can give rise to convulsions in epileptic

Warnings: Flumazenil is used with neuromuscular blocking agents, it should not be injected until the effects of neuromuscular blockade have been fully reversed. Rapid injection of Flumazenil should be avoided in patients with long-term exposure to benzodiazepines ending at any time within weeks preceding Flumazenil administration as it may produce withdrawal symptoms, including agitation, anxiety, emotional lability as well as mild confusion and sensory distortions. Flumazenil is not recommended either as a treatment for benzodiazepine dependence or for the management of protracted benzodiazepine abstinence syndromes.

Effects on the ability to drive and use machinery:

Patients should be warned against engaging in hazardous activities requiring complete mental alertness (such as operating dangerous machinery or driving a motor vehicle) during the first 24 hours after administration since the effect on the originally ingested or administered benzodiazepine (for example, sedation) may occur.

Use in Children: No data available.

INTERACTION WITH OTHER MEDICATIONS: Flumazenil blocks the central effects of benzodiazepines by competitive interaction at the receptor level; the effects of non-benzodiazepines against benzodiazepine receptors, such as zopiclone, triazolopyridazines and others, are also blocked by flumazenil. The pharmacokinetics of benzodiazepines against are unaltered in the presence of flumazenil and vice versa.

USE IN PREGNANCY & LACTATION: **Use in Pregnancy:** Although in-vitro and animal studies using high doses of flumazenil have not shown evidence of mutagenicity, teratogenicity or impairment of fertility, the safety of flumazenil in human pregnancy has not been established. Therefore, the benefits of drug therapy during pregnancy should be weighed against possible risk to the fetus.

Use in Lactation: Parenteral administration of flumazenil in emergencies is not contraindicated during lactation.

SIDE EFFECTS: Flumazenil was systemically and locally well tolerated. Nausea and/or vomiting was reported in clinical trials with Flumazenil. This occurred more frequently when Flumazenil was given as a single high dose to reverse anaesthesia and when opioids and other anaesthetic agents were used as a component of the anaesthesia. These reactions occurred rarely in volunteer studies or when benzodiazepines alone were used for sedation.

Complaints such as feelings of anxiety, palpitations and fear have been infrequently observed after rapid injection of flumazenil. These undesirable effects usually do not necessitate special treatment.

Seizures have been reported in patients known to suffer from epilepsy or severe hepatic impairment, particularly after long-term treatment with benzodiazepines or in cases of mixed-drug overdose.

In cases of mixed-drug overdose, particularly with cyclic antidepressants, toxic effects (such as convulsions and cardiac dysrhythmias) may emerge with the reversal of benzodiazepine effects by Flumazenil.

Withdrawal symptoms may occur following rapid injection of Flumazenil in patients with long-term exposure to benzodiazepines ending at any time within the weeks preceding Flumazenil administration.

Flumazenil has been reported to provoke panic attacks in patients with a history of panic disorders.

INCOMPATIBILITIES:

This medicinal product must not be mixed with other medicinal product except for those mentioned in recommended dosage.

SYMPTOMS AND TREATMENT OF OVERDOSE: Large intravenous dose of Flumazenil, when administered to healthy normal volunteers in the absence of a benzodiazepine agonist, produced no serious adverse reactions, severe signs or symptoms, or clinically significant laboratory test abnormalities. In clinical studies, most adverse reactions to Flumazenil were an extension of the pharmacologic effects of the drug in reversing benzodiazepine effects. Reversal with an excessively high dose of Flumazenil may produce anxiety, agitation, increased muscle tone, hyperesthesia and possibly convulsions.

STORAGE CONDITIONS: Store below 30°C. Protect from light and freezing. Keep out of reach of children. JAUHKA DARIPADA KANAK-KANAK.

PACK SIZE: Box of 50's and 10's ampoules x 5ml.

SHELF LIFE: Please refer to outer package

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

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